

REST API

Deb Angst

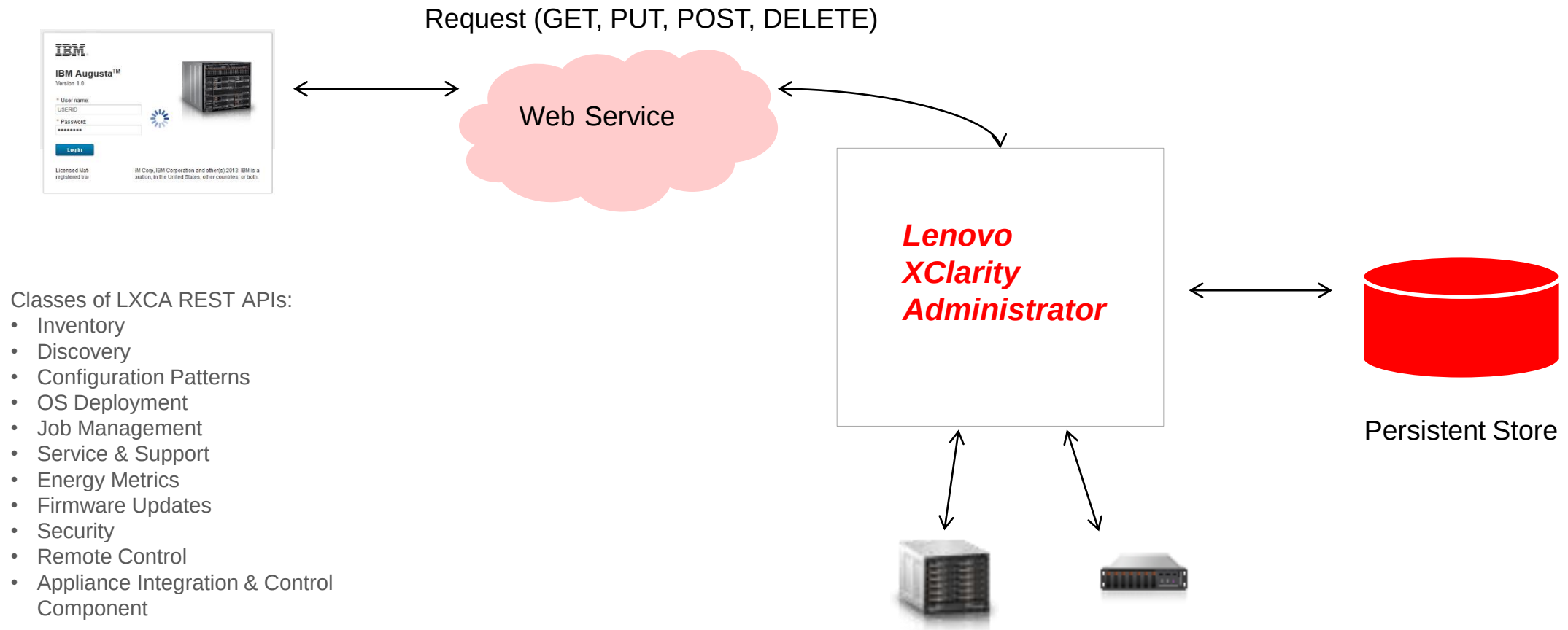


What is REST

- Representational State Transfer (REST)
- Stateless, client-server architecture that can use HTTP protocol
- Resource-based design identified by URIs
- HTTP verbs: GET, PUT, POST, DELETE
- JSON representation transferred between client and server
- HTTP response (state, body)

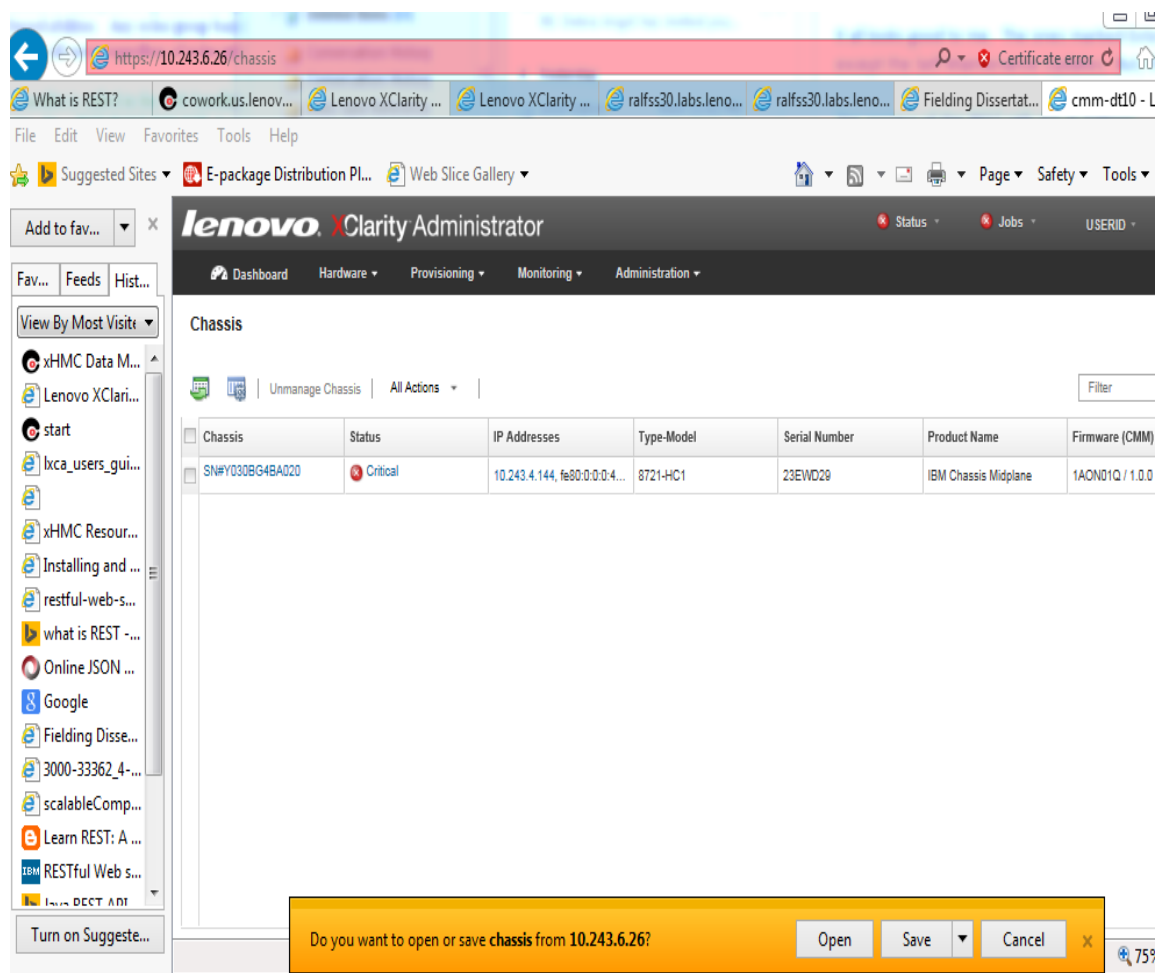
REST in LXCA

- Client applications use the set of REST APIs to access LXCA data and services



LXCA REST API Example

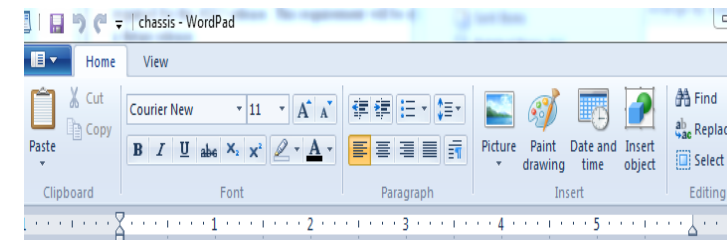
■ GET /chassis



The screenshot shows the Lenovo XClarity Administrator web interface. The browser address bar displays `https://10.243.6.26/chassis`. The interface includes a navigation menu with options like Dashboard, Hardware, Provisioning, Monitoring, and Administration. The main content area shows a table of chassis information.

Chassis	Status	IP Addresses	Type-Model	Serial Number	Product Name	Firmware (CMM)
SN#Y030B04BA020	Critical	10.243.4.144, fe80:0:0:4...	8721-HC1	23EWD29	IBM Chassis Midplane	1A0N01Q / 1.0.0

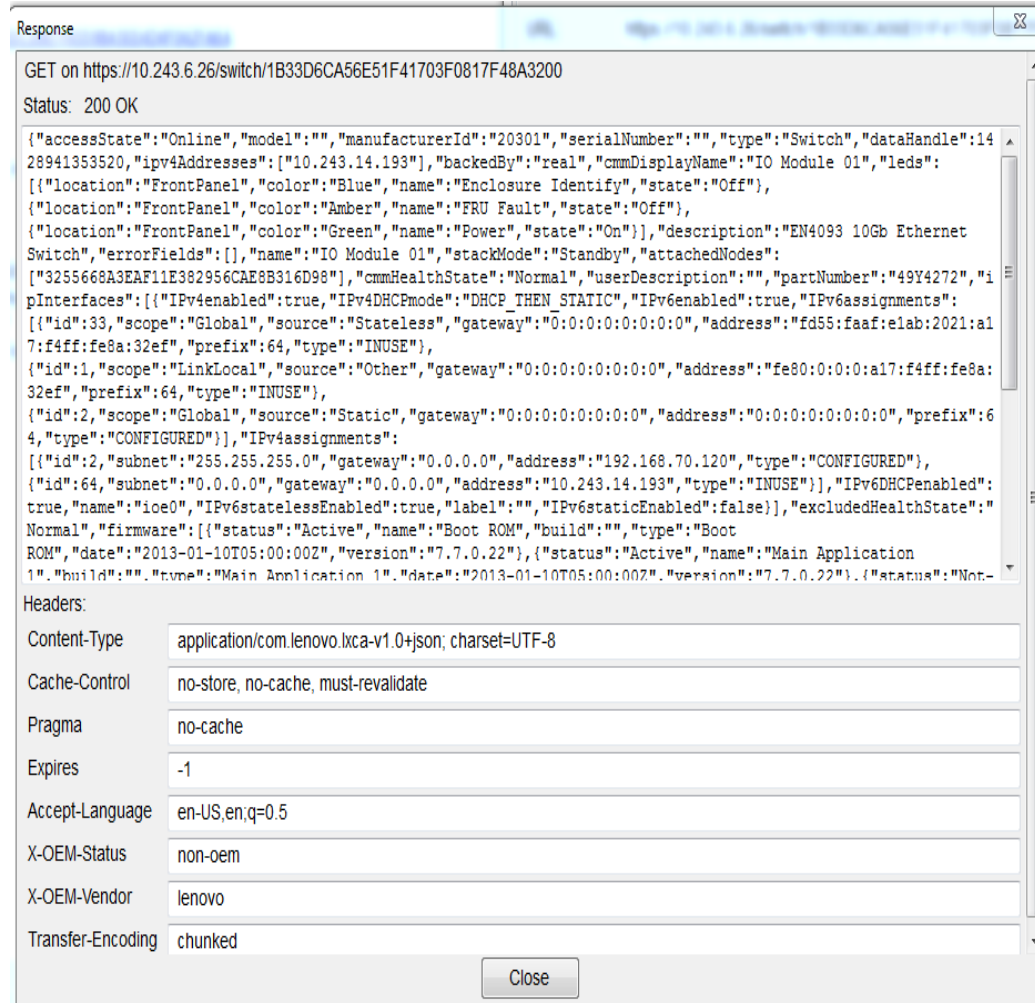
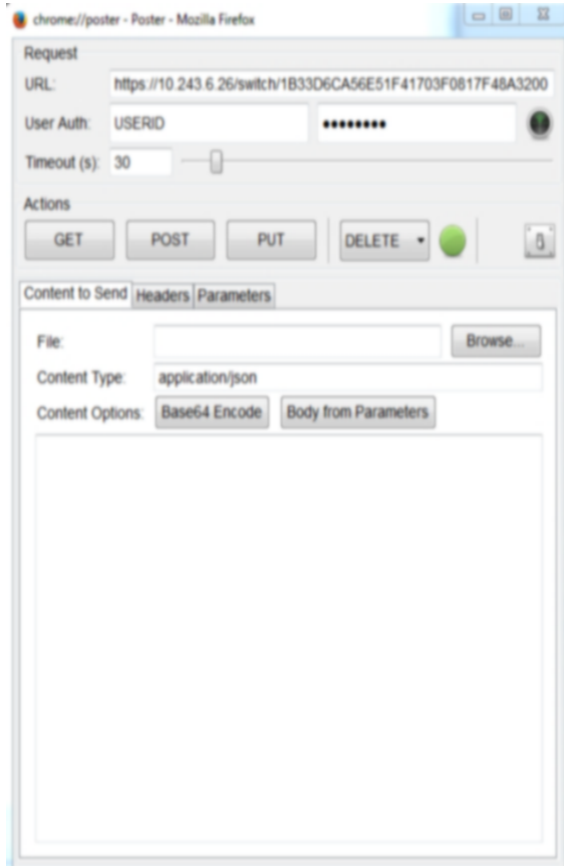
A modal dialog is open at the bottom of the screen with the text: "Do you want to open or save chassis from 10.243.6.26?". The dialog has three buttons: "Open", "Save", and "Cancel".



```
{
  "chassisList": [
    {
      "accessState": "Online",
      "powerSupplySlots": 6,
      "manufacturerId": "20301",
      "location": {
        "lowestRackUnit": 0,
        "location": "",
        "rack": "",
        "room": "",
        "serialNumber": "23EWD29",
        "type": "Chassis",
        "fanSlots": 10,
        "height": 10,
        "switchSlots": 4,
        "passThroughModules": [],
        "leds": [
          {
            "location": "FrontPanel",
            "color": "Blue",
            "name": "Location",
            "state": "Off"
          },
          {
            "location": "FrontPanel",
            "color": "Amber",
            "name": "FAULT",
            "state": "On"
          },
          {
            "location": "FrontPanel",
            "color": "Amber",
            "name": "Information",
            "state": "On"
          }
        ],
        "description": "Lenovo Flex System Chassis",
        "errorFields": [],
        "mgmtProcIpAddress": "10.243.4.144",
        "activationKeys": [],
        "status": {
          "message": "MANAGED",
          "name": "MANAGED",
          "hostname": "MM40F2E9BF5A34",
          "powerSupplies": [
            {
              "model": "",
              "manufacturerId": "20301",
              "inputVoltageIsAC": true,
              "serialNumber": "",
              "type": "Power Supply",
              "inputVoltageMin": 220,
              "dataHandle": 0,
              "cmmDisplayName": "Power Supply 01",
              "leds": [
                {
                  "location": "FRU",
                  "color": "Green",
                  "name": "IN",
                  "state": "On"
                },
                {
                  "location": "FRU",
                  "color": "Green",
                  "name": "OUT",
                  "state": "On"
                },
                {
                  "location": "FRU",
                  "color": "Amber",
                  "name": "FAULT",
                  "state": "Off"
                }
              ],
              "description": "Power Supply",
              "name": "Power Supply 01",
              "cmmHealthState": "Normal",
              "manufactureDate": "2311",
              "userDescription": "",
              "partNumber": "69Y5801",
              "firmware": {
                "status": "",
                "name": "Power Supply Firmware",
                "role": "",
                "build": "",
                "type": "Power Supply Firmware",
                "date": "",
                "version": "0"
              },
              "machineType": "",
              "hardwareRevision": "5.0",
              "parent": {
                "uuid": "63E29269BB634AB9A610D6F8FCE2B28F",
                "uri": "chassis/63E29269BB634AB9A610D6F8FCE2B28F",
                "FRU": "69Y5817",
                "uri": "powerSupply/6392ACAF973B11E00007000700070007",
                "productId": "303",
                "powerAlloca"
              }
            }
          ]
        }
      }
    }
  ]
}
```

LXCA REST API Example cont'd

■ GET /switch/{UUID}



LXCA REST API Example cont'd

■ PUT /switch/{uuid} – change user description

chrome://poster - Poster - Mozilla Firefox

Request

URL:

User Auth:

Timeout (s):

Actions

GET POST PUT DELETE

Content to Send Headers Parameters

File: Browse...

Content Type:

Content Options:

(userDescription:"My 01 Switch")

Response

PUT on https://10.243.6.26/switch/1B33D6CA56E51F41703F0817F48A3200

Status: 200 OK

Headers:

Content-Type	application/com.lenovo.lxca-v1.0+json; charset=UTF-
Cache-Control	no-store, no-cache, must-revalidate
Pragma	no-cache
Expires	Thu, 01-Jan-1970 00:00:00 GMT
Accept-Language	en-US,en;q=0.5
X-OEM-Status	non-oem
X-OEM-Vendor	lenovo
Set-Cookie	LXCASESSID=qilh4s928ijochv4i0y1nq9u;Path=/switc

Close

Response

GET on https://10.243.6.26/switch/1B33D6CA56E51F41703F0817F48A3200

Status: 200 OK

```
{
  "accessState": "Online",
  "model": "",
  "manufacturerId": "20301",
  "serialNumber": "",
  "type": "Switch",
  "dataHandle": "1428943497522",
  "ipv4Addresses": [
    {
      "location": "FrontPanel",
      "color": "Blue",
      "name": "Enclosure Identify",
      "state": "Off"
    },
    {
      "location": "FrontPanel",
      "color": "Amber",
      "name": "FRU Fault",
      "state": "Off"
    },
    {
      "location": "FrontPanel",
      "color": "Green",
      "name": "Power",
      "state": "On"
    }
  ],
  "description": "EN4093 10Gb Ethernet Switch",
  "errorFields": [
    {
      "name": "IO Module 01",
      "stackMode": "Standby",
      "attachedNodes": [
        {
          "id": "33",
          "scope": "Global",
          "source": "Stateless",
          "gateway": "0:0:0:0:0:0:0:0",
          "address": "fd55:faaf:elab:2021:a17:f4ff:fe8a:32ef",
          "prefix": 64,
          "type": "INUSE"
        },
        {
          "id": "1",
          "scope": "LinkLocal",
          "source": "Other",
          "gateway": "0:0:0:0:0:0:0:0",
          "address": "fe80:0:0:0:a17:f4ff:fe8a:32ef",
          "prefix": 64,
          "type": "INUSE"
        },
        {
          "id": "2",
          "scope": "Global",
          "source": "Static",
          "gateway": "0:0:0:0:0:0:0:0",
          "address": "0:0:0:0:0:0:0:0",
          "prefix": 64,
          "type": "CONFIGURED"
        },
        {
          "id": "2",
          "subnet": "255.255.255.0",
          "gateway": "0.0.0.0",
          "address": "192.168.70.120",
          "type": "CONFIGURED"
        },
        {
          "id": "64",
          "subnet": "0.0.0.0",
          "gateway": "0.0.0.0",
          "address": "10.243.14.193",
          "type": "INUSE"
        }
      ]
    }
  ],
  "cmmHealthState": "Normal",
  "userDescription": "My 01 Switch",
  "partNumber": "49Y4272",
  "ipInterfaces": [
    {
      "id": "33",
      "scope": "Global",
      "source": "Stateless",
      "gateway": "0:0:0:0:0:0:0:0",
      "address": "fd55:faaf:elab:2021:a17:f4ff:fe8a:32ef",
      "prefix": 64,
      "type": "INUSE"
    },
    {
      "id": "1",
      "scope": "LinkLocal",
      "source": "Other",
      "gateway": "0:0:0:0:0:0:0:0",
      "address": "fe80:0:0:0:a17:f4ff:fe8a:32ef",
      "prefix": 64,
      "type": "INUSE"
    },
    {
      "id": "2",
      "scope": "Global",
      "source": "Static",
      "gateway": "0:0:0:0:0:0:0:0",
      "address": "0:0:0:0:0:0:0:0",
      "prefix": 64,
      "type": "CONFIGURED"
    },
    {
      "id": "2",
      "subnet": "255.255.255.0",
      "gateway": "0.0.0.0",
      "address": "192.168.70.120",
      "type": "CONFIGURED"
    },
    {
      "id": "64",
      "subnet": "0.0.0.0",
      "gateway": "0.0.0.0",
      "address": "10.243.14.193",
      "type": "INUSE"
    }
  ]
}
```

Headers:

Content-Type	application/com.lenovo.lxca-v1.0+json; charset=UTF-8
Cache-Control	no-store, no-cache, must-revalidate
Pragma	no-cache
Expires	-1
Accept-Language	en-US,en;q=0.5
X-OEM-Status	non-oem
X-OEM-Vendor	lenovo
Transfer-Encoding	chunked

Close

LXCA REST API Example cont'd

- PUT /switch/{uuid} – change power state

The screenshot shows a REST client interface with a 'Request' tab on the left and a 'Response' tab on the right. The 'Request' tab shows a PUT request to the URL `https://10.243.6.26/switch/1B33D6CA56E51F41703F0817F48A3200` with a 'User Auth' field containing 'USERID' and a password mask. The 'Content Type' is set to 'application/json' and the 'Content Options' are 'Base64 Encode' and 'Body from Parameters'. The 'Content to Send' field contains `{"powerState": "powerOff"}`. The 'Response' tab shows a 200 OK status and a list of headers including 'Content-Type', 'Cache-Control', 'Pragma', 'Expires', 'Accept-Language', 'X-OEM-Status', 'X-OEM-Vendor', 'Set-Cookie', and 'Location'.

Request

URL: `https://10.243.6.26/switch/1B33D6CA56E51F41703F0817F48A3200`

User Auth: USERID [password mask]

Timeout (s): 30

Actions: GET POST PUT DELETE

Content to Send Headers Parameters

File: [Browse...]

Content Type: application/json

Content Options: Base64 Encode Body from Parameters

`{"powerState": "powerOff"}`

Response

PUT on `https://10.243.6.26/switch/1B33D6CA56E51F41703F0817F48A3200`

Status: 200 OK

Headers:

- Content-Type: application/com.lenovo.lxca-v1.0+json; charset=UTF-
- Cache-Control: no-store, no-cache, must-revalidate
- Pragma: no-cache
- Expires: Thu, 01-Jan-1970 00:00:00 GMT
- Accept-Language: en-US,en;q=0.5
- X-OEM-Status: non-oem
- X-OEM-Vendor: lenovo
- Set-Cookie: LXCASESSID=fnygb5atpzo4jomd7iz5sjo; Path=/switc
- Location: /jobs/15

Switches

The screenshot shows a table titled 'Switches' with columns: Switch, Status, Power, Chassis, and Bay. The first row is for 'IO Module 01' with a status of 'Normal' and a power state of 'On'. The 'Power' column is circled in red.

Switch	Status	Power	Chassis	Bay
IO Module 01	Normal	On	SN#Y030BG4BA020	1

Switches

The screenshot shows a table titled 'Switches' with columns: Switch, Status, Power, Chassis, and Bay. The first row is for 'IO Module 01' with a status of 'Normal' and a power state of 'Off'. The 'Power' column is circled in red.

Switch	Status	Power	Chassis	Bay
IO Module 01	Normal	Off	SN#Y030BG4BA020	1

LXCA REST API Example cont'd

■ PUT /switch/{uuid}

Response

PUT on https://10.243.6.26/switch/1B33D6CA56E51F41703F0817F48A3200

Status: 200 OK

Headers:

Content-Type	application/com.lenovo.lxca-v1.0+json; charset=UTF-
Cache-Control	no-store, no-cache, must-revalidate
Pragma	no-cache
Expires	Thu, 01-Jan-1970 00:00:00 GMT
Accept-Language	en-US,en;q=0.5
X-OEM-Status	non-oem
X-OEM-Vendor	lenovo
Set-Cookie	LXCASessionID=1b8r5d370g5f16f5vikffhoh7;Path=/swi
Location	/jobs/16

Close

Response

GET on https://10.243.6.26/jobs/16

Status: 200 OK

```
{
  "startTime": "2015-04-13T16:57:26Z",
  "id": 16,
  "category": "POWER",
  "messageID": "0512_LONG",
  "createdBy": "SYSMGR_XRPYLX4P",
  "cancelledBy": "",
  "status": {
    "substatus": [
      {
        "messageID": "0514_LONG",
        "status": {
          "percentage": "100",
          "state": "Complete",
          "messageDisplay": "Power management job for IO Module 01 has completed successfully.",
          "messageBundle": "com.lenovo.lxca.discovery.bundle.rest.messages",
          "uuid": "1B33D6CA56E51F41703F0817F48A3200",
          "messageParameters": [
            "IO Module 01"
          ]
        },
        "percentage": "100",
        "description": [
          {
            "messageID": "0516_LONG",
            "messageTime": "2015-04-13T12:57:31Z",
            "messageDisplay": "Power management job has finished.",
            "messageBundle": "com.lenovo.lxca.discovery.bundle.rest.messages",
            "state": "Complete",
            "messageDisplay": "Power management job for IO Module 01",
            "messageBundle": "com.lenovo.lxca.discovery.bundle.rest.messages",
            "uuid": "1B33D6CA56E51F41703F0817F48A3200",
            "endTime": "2015-04-13T16:57:31Z",
            "messageParameters": [
              "IO Module 01"
            ],
            "typeId": 0
          }
        ]
      }
    ]
  }
}
```

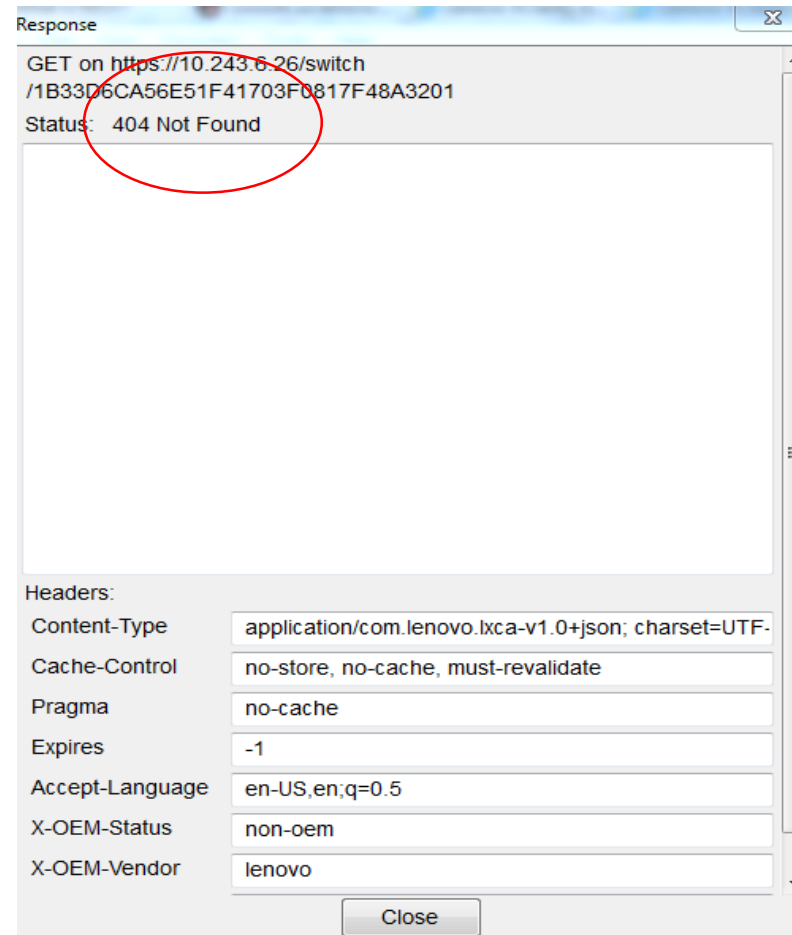
Headers:

Content-Type	application/com.lenovo.lxca-v1.0+json; charset=UTF-
Cache-Control	no-store, no-cache, must-revalidate
Pragma	no-cache
Expires	-1
Accept-Language	en-US,en;q=0.5
X-OEM-Status	non-oem
X-OEM-Vendor	lenovo
Transfer-Encoding	chunked

Close

LXCA REST API Example cont'd

- GET /switch/{uuid} – unknown UUID specified



- GA1 LXCA REST API documentation location:

- Flex System Information Center: <http://pic.dhe.ibm.com/infocenter/flexsys/information/index.jsp>

THANK YOU GRACIAS
MERCI DANKE GRAZIAS 謝謝
OBRIGADO ありがとう DANK TAKK BEDANKT СПАСИБО
DAKUJEM